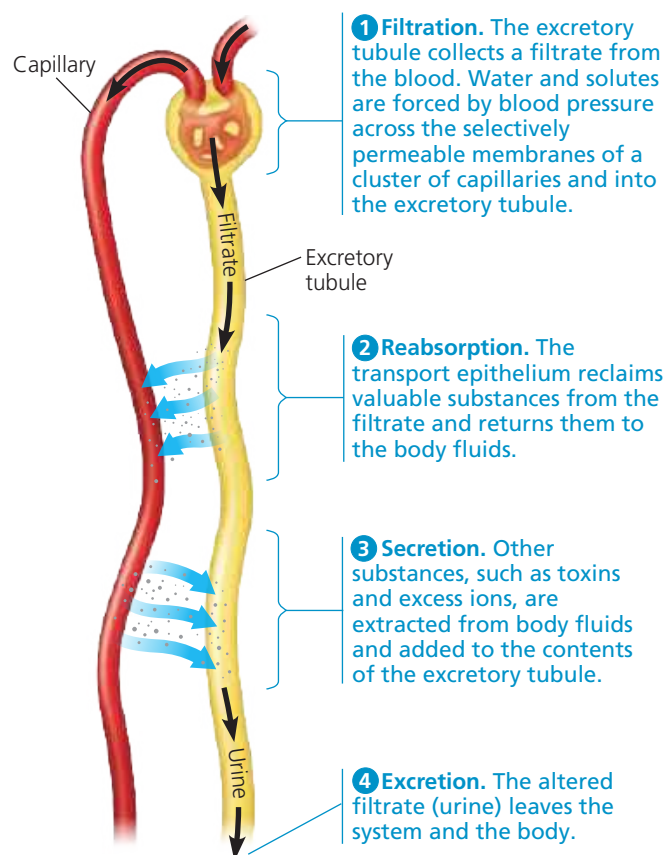


▼ **Figure 44.8 Key steps of excretory system function: an overview.** Most excretory systems produce a filtrate by pressure-filtering body fluids and then modify the filtrate's contents. This diagram is modeled after the vertebrate excretory system.



in contact with the selectively permeable membrane of a transport epithelium (**Figure 44.8**). In most cases, hydrostatic pressure (blood pressure in many animals) drives a process of **filtration**. Cells, as well as proteins and other large molecules, cannot cross the epithelial membrane and remain in the body fluid. In contrast, water and small solutes, such as salts, sugars, amino acids, and nitrogenous wastes, cross the membrane, forming a solution called the **filtrate**.

The filtrate is converted to a waste fluid by the specific transport of materials into or out of the filtrate. The process of selective **reabsorption** recovers useful molecules and water from the filtrate and returns them to the body fluid. Valuable solutes—including glucose, certain salts, vitamins, hormones, and amino acids—are reabsorbed by active transport. Nonessential solutes and wastes are left in the filtrate or are added to it by selective **secretion**, which also occurs by active transport. The pumping of various solutes in turn determines whether water moves by osmosis into or out of the filtrate. In the last step—excretion—the processed filtrate containing nitrogenous wastes is released from the body as a fluid waste called urine.

Survey of Excretory Systems

The systems that perform the basic excretory functions outlined in **Figure 44.8** vary widely among animal groups. However, they are generally built on a complex network of tubules that provide a large surface area for the exchange of water and solutes, including nitrogenous wastes. We'll examine the excretory systems of flatworms, earthworms, insects, and vertebrates as examples of evolutionary variations on tubule networks.

Protonephridia

As illustrated in **Figure 44.9**, flatworms (phylum Platyhelminthes), which lack a coelom or body cavity, have excretory systems called **protonephridia** (singular, *protonephridium*). Protonephridia consist of a network of dead-end tubules that branch throughout the body. Cellular units called flame bulbs cap each branch. Each flame bulb, consisting of a tubule cell and a cap cell, has a tuft of cilia projecting into the tubule.

During filtration, the beating of the cilia draws water and solutes from the interstitial fluid through the flame bulb, releasing filtrate into the tubule network. (The name *flame bulb* derives from the moving cilia's resemblance to a flickering flame.) The processed filtrate moves outward through the tubules and empties as urine via external openings. Because the urine excreted by freshwater flatworms is low in solutes, its production helps to balance the osmotic uptake of water from the environment.

▼ **Figure 44.9 Protonephridia in a planarian.**

